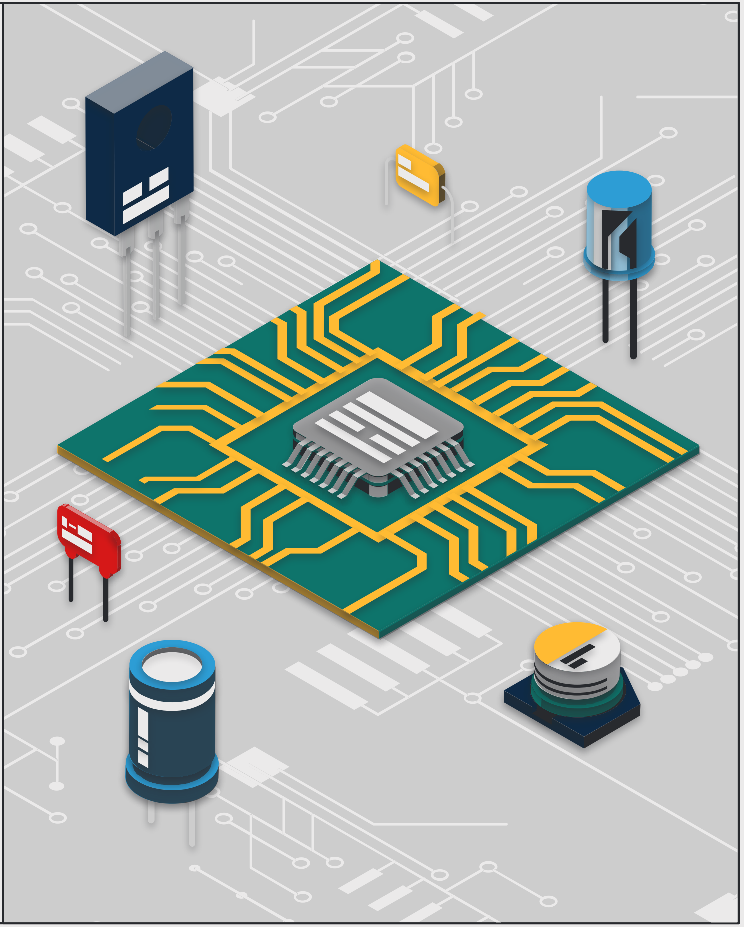
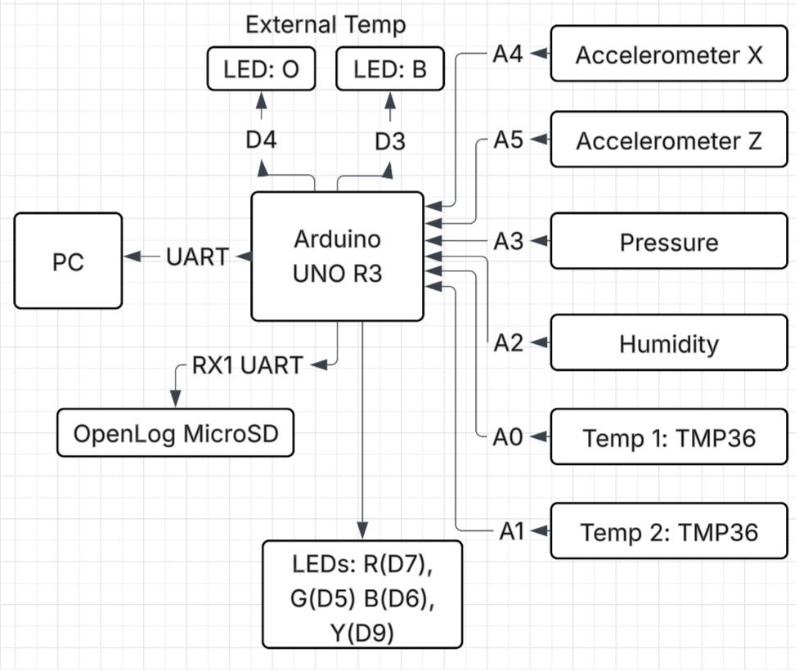
# NASA Colorado Space Grant Consortium

Arduino Shield - Aidan Donnellan



# Current Architecture

Block diagram for current Arduino UNO R3 Hat used for NASA CO. Space Grant Consortium




Block Diagram made using LucidChart

# Current Component Selection

| Component | Purpose | Power Consumption (mA) | Weight (g) |
|---|---|---|---|
| Arduino UNO R3 | Brains | ~20-75 | ~25 |
| TMP36 | Internal and External Temperature | <.5 (data sheet) → <1 | ~0.2 → 0.4 |
| HIH-4030 | Humidity Sensor | .2 (link) | ~1 |
| ADXL335 | Accelerometer | .35 (data sheet) | ~4.54 |
| SSCDANN015 PD2A5 | 8-pin Pressure Sensor | <10, avg. 3 (datasheet) | ~1 |
| LED's | STD. 5mm (O, (2)B, R, G, Y) | 16-18 (link) | ~0.3 |
| OpenLog MicroSD | Serial Logger | 2-3 (idle), 10-20 (powered) (link) | 3 |
| PCB | - | - | ~15 |
| Header Pins | - | - | ~1.8 |
| **Total Power Consumption:** 119.35 (max) | | **Total Weight:** ~52.04 (not considering battery) | |

# Feasibility Assessment – 1

| **Component** | **Feasibility / Alternative** |
|---|---|
| Arduino UNO R3 | Is bulky and heavy, alternatives such as the Nano Every or Teensy 4.x provide lightweight solutions with same, if not better, logic. |
| HIH-4030 | Reliable, however, condensation or moisture exposure can impact performance. |
| SSCDANN015PD2A5 | High quality, expensive pressure sensor. Good for stratospheric pressure accuracy |
| ADXL335 | Low power, susceptible to signal noise |
| OpenLog | Necessary for data collection, other open log models exist if a specific device is not available. |
| TMP36 | Good for sub-stratosphere testing, not suited for stratosphere temperatures. |

# Feasibility Assessment – 2

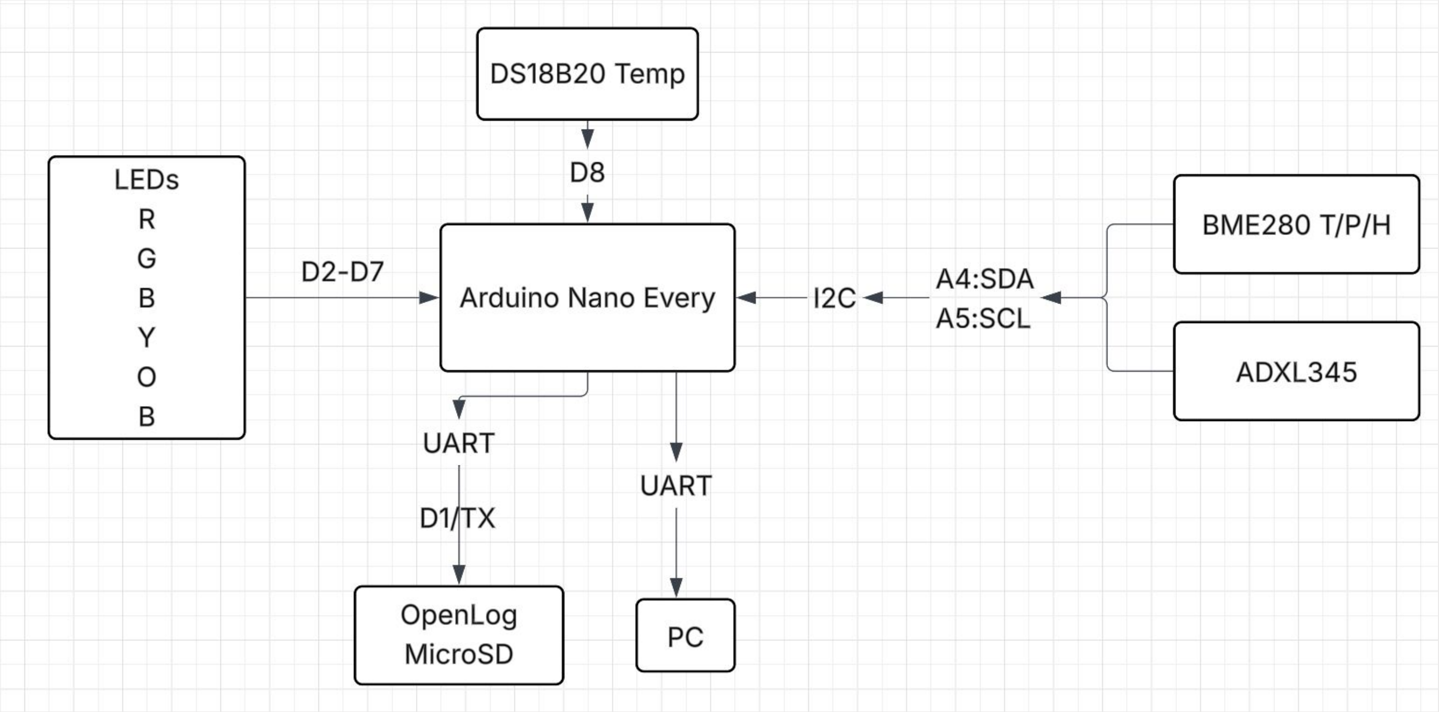
Component Feasibility Assessment: The primary feasibility issues lies not within components being unavailable, rather the thermal and power limits of certain components. At stratospheric temperatures, certain components such as the TMP36 are only rated down to -40 degrees celsius. Once the balloon passes into the stratosphere, the sensor is highly likely to flatline or give incorrect data, making this component in specific unsuitable for use.

Component Unavailability: Most components used within the project have strong alternatives with similar ratings. However, weight and cost play a substantial role in component selection.

Overall: The current system is well suited for low-altitude testing; however, for NASA compliance and full scale launch, certain components should be reconsidered.

Pre-Phase 2 Plan: Design a PCB utilizing an Arduino Nano Every and BME280 temperature/humidity/barometric pressure sensor for decreased payload and form factor.

# Revised Architecture




Block Diagram made using LucidChart

# Component Selection

| **Component** | **Purpose** | **Power Consumption (mA)** | **Weight (g)** | **Cost (USD)** |
|---|---|---|---|---|
| Arduino Nano Every | Brains | ~18 | 5 | 13 |
| DS18B20 | External Temp | 1 | 1 | 15 |
| BME280 | Internal Temp, Pressure, Humidity | 0.1 | 15 | 10 |
| ADXL345 | Accelerometer | 0.1 | 2 | 18 |
| LED’s | STD. 5mm (O, (2)B, R, G, Y) | 16-18 (link) | 2 | 18 |
| OpenLog MicroSD | Serial Logger | 2-3 (idle), 10-20 (powered) (link) | 2 | 5 |
| PCB | - | - | 20 | 30 |
| Header Pins | - | - | | |
| **Total Power Consumption:** 135 (max), ~36 (status) | | **Total Weight:** ~47 (not considering battery) | | **Total Cost:** 109 |

# Component Specifications

**Arduino Nano Every**

ATMega4809

- AVR CPU up to 20MHz
- 48 kB Flash
- 6 kB SRAM
- 256B EEPROM

ATSAMD11D14A

- Arm Cortex M0+ at up to 48MHz
- 16KB Flash
- 4KB SRAM
- 12Mbps USB 2.0 Interface

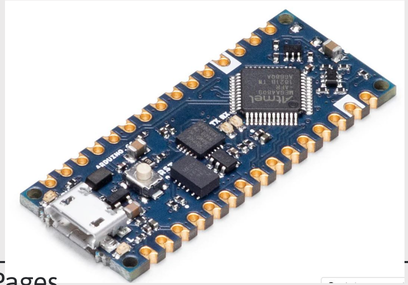
- **Peripherals**
  - 1x 16-bit Timer/Counter with a dedicated period register and 3x compare channels
  - 4x 16-bit Timer/Counter with input capture
  - 1x 16-bit Real-Time Counter (RTC) running from an external crystal or an internal RC oscillator
  - 4x USART with fractional baud rate generator, auto-baud, and start-of-frame detection
  - 1x Master/slave Serial Peripheral Interface (SPI)
  - 1x Dual mode Master/Slave TWI with dual address match 6x 16 bit Timers (1 dedicated to RTC)
  - Event System for CPU independent and predictable inter-peripheral signaling
  - Configurable Custom Logic (CCL) with up to four programmable Look-up Tables (LUT)
  - 1x Analog Comparator (AC) with a scalable reference input
  - Watchdog Timer with Window mode, with separate on-chip oscillator
  - External interrupt on all general purpose pins

Arduino Nano Every Data Sheet: ABX00028

# Why Arduino Nano Every

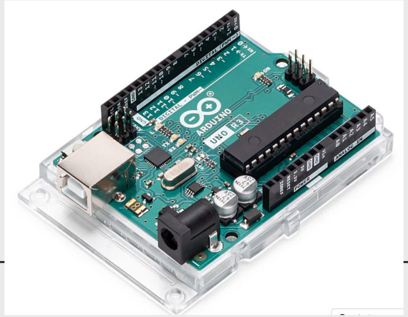
**<u>Arduino Nano Every</u>**

- Smaller Form Factor
- ATMega4809
    - 48 KB Flash
    - 6 KB SRAM
    - 256 Bytes EEPROM
    - CPU Speed: 20 MHz
    - Hardware UARTS: 4
    - ADC Channels: 16 (10-bit)



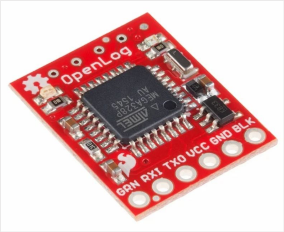
**<u>Arduino UNO</u>**

- Larger Form Factor
- ATMega328P
    - 32 KB Flash
    - 2KB SRAM
    - 1KB EEPROMs
    - CPU Speed: 20 MHz
    - Hardware UARTS: 1
    - ADC Channels: 8 (10-bit)



Photos from Product Pages

# Component Specifications




SparkFun BME280


- Operating Voltage: 1.71V-3.6V
- Temp Range: -40 C to +85 C
- Humidity: 0% to 100%
- Pressure 300 hPa to 1100 hPa
- Altitude: 0 to 30,000 ft.
- Interfacing: I2C (Address 0x77) up to 3.4MHz

SparkFun ADXL345

- Operating Voltage: 2.0V - 3.6V
- Resolution: up to 13-bit
- Output Rate: 0.1 to 3200 Hz
- Interfacing: I2C (Address 0x53) and SPI
- Operating Temp: -40 C to + 85 C


SparkFun OpenLog

- Operating Voltage: 3.3V to 5V
- Baud: 300 to 115,200 bps
- Storge: microSD up to 32GB
- Processor: ATMega382P

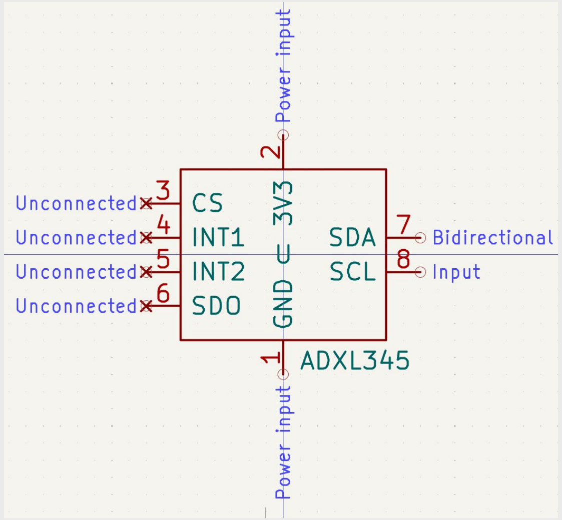
SparkFun DS18B20 Temp

- Operating Voltage: 3.0V to 5.5V
- Operating Temp: -55 C to +125 C
- Resolution: 9 to 12 bits
- Communication: 1 wire interface
- 1m cable length

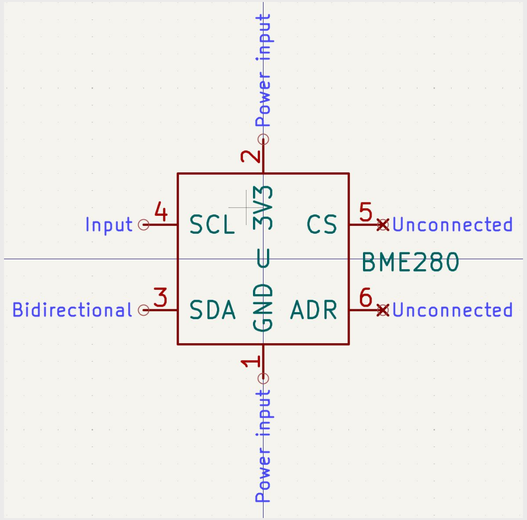
Photos from Product Pages

# Custom Symbols

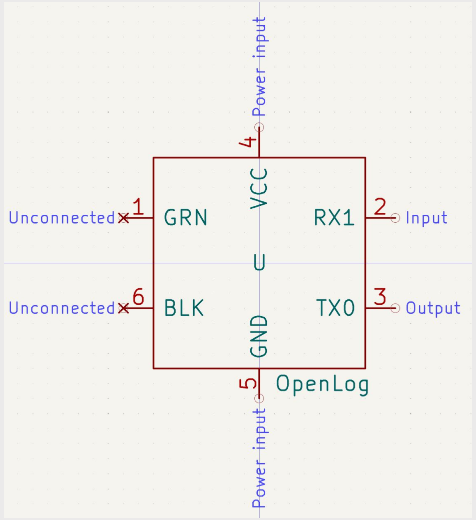
Creating Custom Symbols Using Symbol Editor




ADXL345 Module



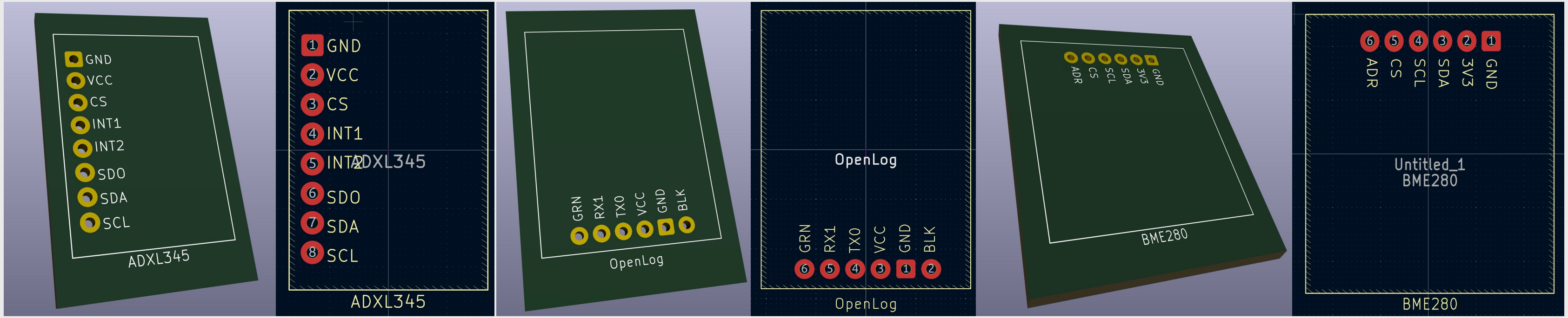

BME280 Module




OpenLog Module

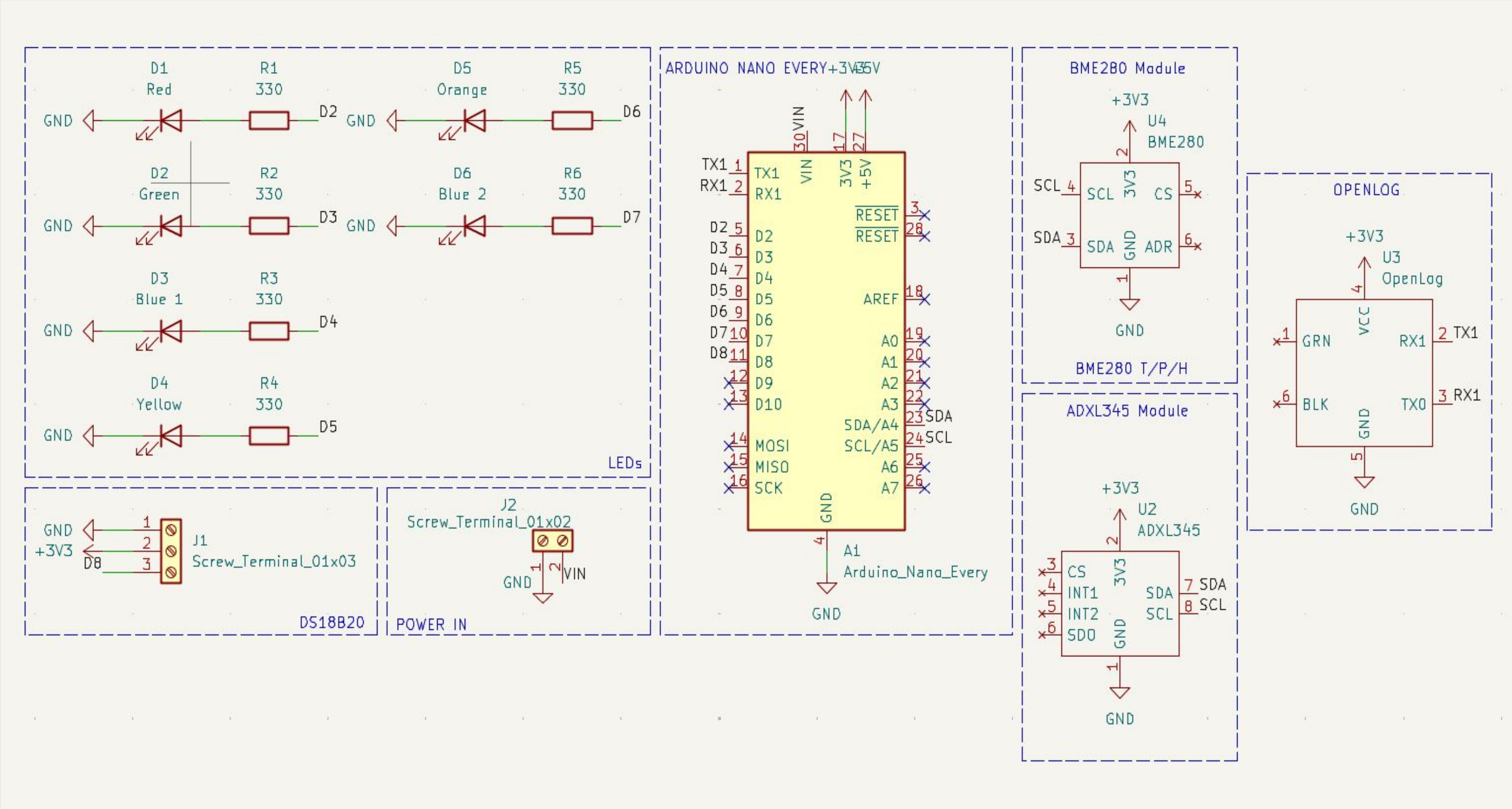
# Custom Footprints

## Creating Custom Footprints Using Footprint Editor and Footprint Assignment



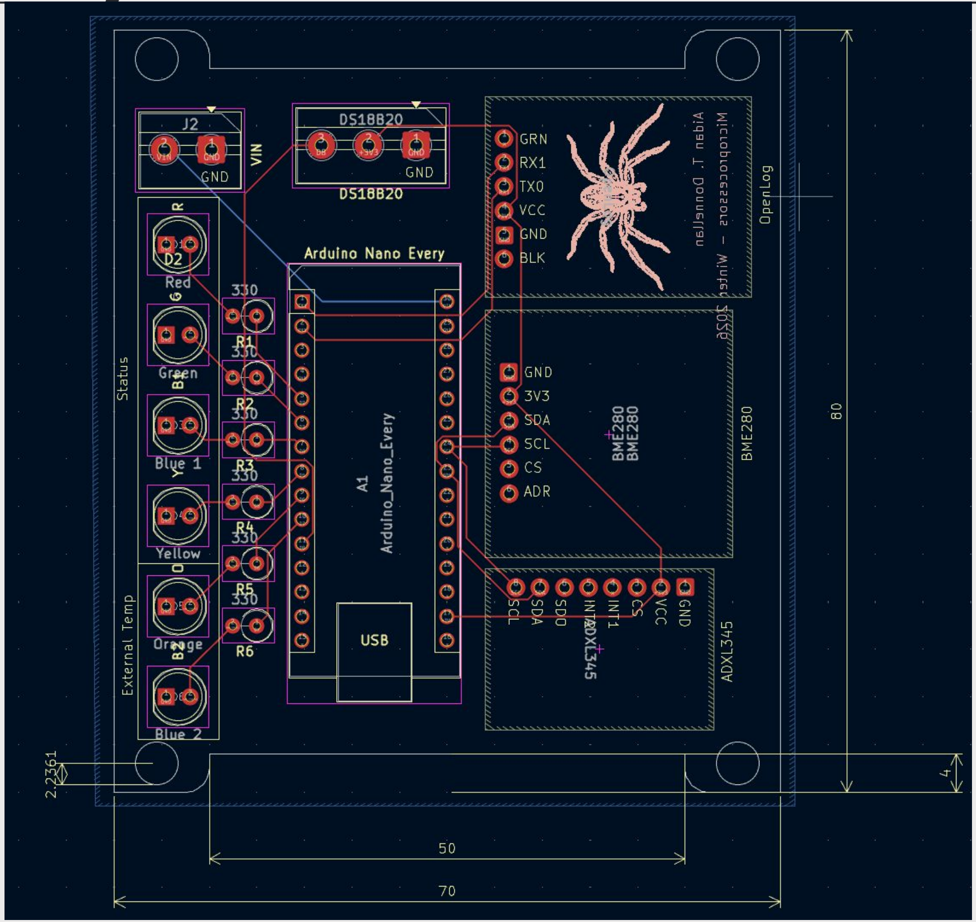

```
 1        A1 - Arduino_Nano_Every : Module:Arduino_Nano
 2        D1 -              Red : LED_THT:LED_D5.0mm
 3        D2 -            Green : LED_THT:LED_D5.0mm
 4        D3 -           Blue 1 : LED_THT:LED_D5.0mm
 5        D4 -           Yellow : LED_THT:LED_D5.0mm
 6        D5 -           Orange : LED_THT:LED_D5.0mm
 7        D6 -           Blue 2 : LED_THT:LED_D5.0mm
 8        J1 - Screw_Terminal_01x03 : TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-03P_1x03_P5.00mm
 9        R1 -              330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
10        R2 -              330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
11        R3 -              330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
12        R4 -              330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
13        R5 -              330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
14        R6 -              330 : Resistor_THT:R_Axial_DIN0309_L9.0mm_D3.2mm_P2.54mm_Vertical
15        U1 -           BME280 : BME280:BME280_fp
16        U2 -          ADXL345 : ADXL345:ADXL_fp
17        U3 -          OpenLog : OpenLog:OpenLog_fp
```

# Schematic

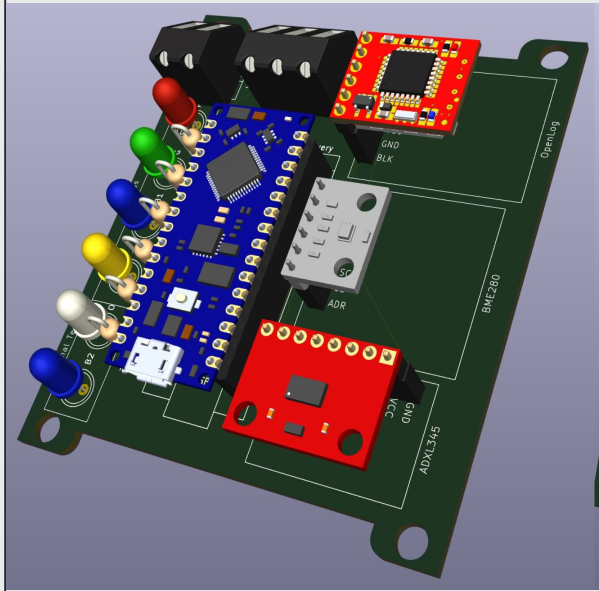


LEDs
D1 Red R1 330 D2
D2 Green R2 330 D3
D3 Blue 1 R3 330 D4
D4 Yellow R4 330 D5
D5 Orange R5 330 D6
D6 Blue 2 R6 330 D7
GND
ARDUINO NANO EVERY
+3V3
+5V
A1
Arduino_Nano_Every
TX1 RX1 D2 D3 D4 D5 D6 D7 D8 D9 D10 MOSI MISO SCK
VIN 3V3 +5V
RESET RESET AREF A0 A1 A2 A3 SDA/A4 SCL/A5 A6 A7
SDA SCL
BME280 Module
U4 BME280
SCL SDA 3V3 GND CS ADR
BME280 T/P/H
ADXL345 Module
U2 ADXL345
CS INT1 INT2 SDO 3V3 GND SDA SCL
OPENLOG
U3 OpenLog
GRN BLK VCC GND RX1 TX0
DS18B20
J1 Screw_Terminal_01x03
+3V3 D8
POWER IN
J2 Screw_Terminal_01x02
VIN

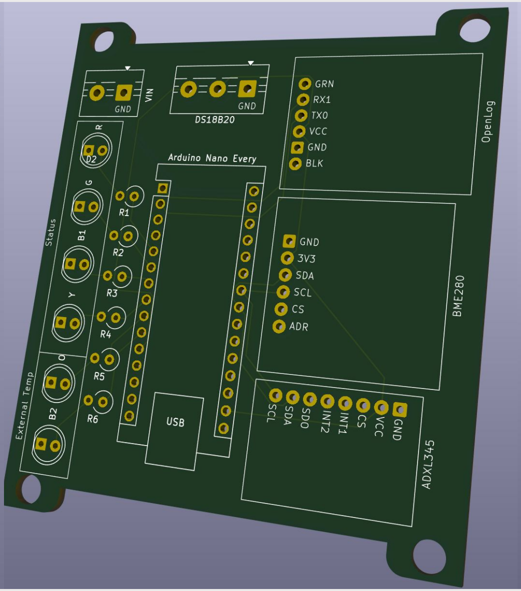
# PCB Assembly



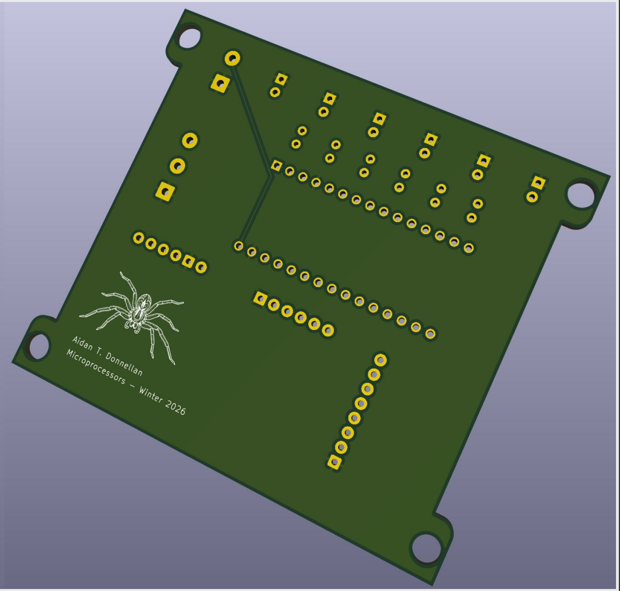
# 3D Render




Component Render




Front




Back

# Current Evaluation for Future Dev

Adding a 2x15 pin socket for extra modularity.

Re-evaluation of certain component choices that better suit stratospheric temperatures.

# Documentation

**KiCAD**

- Arduino Nano Every 3D STP: Arduino Nano Every 3D STP
- SparkFun ADXL345 3D STP: ADXL345 3D STP
- BME280 Module: Lauren Ooghe
- SparkFun OpenLog: Lauren Ooghe
- HTM Workshop Playlist: Playlist

**Reference Material**

- Nasa Space Grant Consortium: High Altitude Balloon

Link to all files can be found here: Donnellan Git

**Data Sheets**

- Arduino Nano Every Data Sheet: ADX00028
- Arduino UNO R3 Data Sheet: AD000066
- SparkFun ADXL345 Data Sheet: ADXL345
- SparkFun BME280 Data Sheet: BME280
- SparkFun OpenLog Product Page: OpenLog
- DS18B20 Temp Data Sheet: DS18B20
- TMP36 Data Sheet: TMP36
- SparkFun HIH 4030 Product Page: HIH 4030
- SparkFun ADXL335 Data Sheet: ADXL335
- Honeywell 8-pin Pressure Sensor: Pressure Sensor